

REECE DICKINSON

Arcata, CA • (707) 498-7150 • reecedickinson06@gmail.com • lettuce.github.io/Portfolio • github.com/Lettuce • linkedin.com/in/reece-dickinson-9178a0419

EDUCATION

Cal Poly Humboldt — Arcata, CA | B.S. in Computer Science

Expected May 2028

- **GPA: 3.94 | Presidential Scholar** (Top Academic Honor, 3.95+ Threshold) | Dean's List (Every Academic Term)
- **Relevant Coursework:** Systems Programming (CS 375), Data Structures (CS 211), Computer Architecture (CS 243), Database Design (CS 325), Web-Apps Using Databases (CS 328), Intro to Data Science (DATA 111), Discrete Math (MATH 253), Biostatistics (STAT 109), Calculus I (MATH 109)

TECHNICAL SKILLS

- **Languages & Frameworks:** C/C++, Python (FastAPI, PyTorch, Pandas, NumPy), Java, SQL (PostgreSQL, Oracle SQL), JavaScript/TypeScript, React.js, Node.js, HTML5, CSS3, R
- **GenAI & Vector Infrastructure:** Full-Stack GenAI, RAG Systems, Vector Embeddings (PGVector), Page-Level Citation Mapping, Local Diffusion Pipelines (Wan 2.2, LTX-2.3, Chroma), ComfyUI Graph Optimization, Real-Time AI StreamDiffusion
- **Systems & Low-Level Infrastructure:** UNIX/Linux Systems Internals, POSIX Multiprocessing ('fork', 'exec', 'wait'), Low-Level I/O Redirections ('<', '>'), Custom Signal Handling ('SIGINT'), Valgrind Memory Profiling, Docker, Git/GitHub, Railway, Supabase, Alembic
- **Certifications & Languages:** Anthropic AI Fluency, Anthropic Claude 101 Certifications | Bilingual (Fluent English, High-Proficiency Conversational Spanish)

TECHNICAL PROJECTS

Full-Stack GenAI Document Copilot & SEC Evaluation Suite

Jun 2026 – Present

Technologies: Python (FastAPI), React, Supabase, PGVector, Alembic, OpenAI APIs

- **Cloud-Native Architecture & Embeddings:** Architected an end-to-end document copilot parsing multi-page SEC financial filings (Apple, NVIDIA, Microsoft, Amazon, Google) with real-time response streaming and PostgreSQL vector embeddings.
- **Trust Contract & Automated RAG Evaluation:** Designed a page-level citation mapping pipeline anchoring AI outputs to exact document coordinates to eliminate hallucinations, and built an automated RAG evaluation harness ('rag_eval_harness.py') achieving a 92.0% pass rate.
- **Beta User Testing & Bug Resolution:** Coordinated hands-on testing with 10 active beta testers to uncover edge-case bugs, optimize prompt responses, and systematically refine query reliability under real-world usage.

Local Generative Media Pipeline & GPU Optimization

Summer 2026 – Present

Technologies: ComfyUI, PyTorch, RTX 4090 Workstation, Node-Graph Engineering

- **GPU Memory Tuning & Source Debugging:** Deployed open-weight image/video diffusion models (Wan 2.2 14B MoE, LTX-2.3 22B) on an RTX 4090 GPU, optimizing FP8 quantization and VRAM efficiency while patching cascading Python dependency breaks via stack trace inspection.
- **Audio-Video Sync & Pipeline Execution:** Engineered an audio-synchronized video pipeline (Wan 2.2 + MMAudio) by resolving frame-rate truncation bugs and fixing file-handoff race conditions using pure in-memory execution node connections.

Custom POSIX C Command Shell Interpreter

Fall 2026 – Present

Technologies: C, POSIX System Calls, Linux Systems Internals, Valgrind, Memory Management

- **Low-Level Multiprocessing & Lifecycle Control:** Engineered a POSIX-compliant C command shell managing parent-child process lifecycles ('fork', 'exec', 'wait') and executing concurrent foreground/background tasks with low-level execution control.
- **Stream Redirection & Zero-Leak Memory Profiling:** Programmed custom file descriptor manipulation routines for I/O stream redirections ('<', '>') and implemented signal handlers ('SIGINT', 'SIGTSTP'), maintaining zero Valgrind memory leaks.

PROFESSIONAL EXPERIENCE

Cal Poly Humboldt Library — Library Scholar Intern — Terraform AI AR Sandbox Project

Feb 2026 – May 2026

- **Real-Time Sensor & AI Pipeline Re-Architecture:** Rebuilt a physical AR topology sandbox from scratch, routing Kinect depth camera inputs into a real-time AI StreamDiffusion pipeline (TouchDesigner, local Python server) running at 60+ FPS to project dynamic 3D terrain contours.
- **Cross-Functional Troubleshooting & MIT Release:** Diagnosed network hops, driver incompatibilities, and frame-rate bottlenecks alongside IT teams; authored technical onboarding runbooks and published the complete codebase under an open-source MIT License.

Ramy Dickinson Construction — Contractor Assistant

Summer 2023 – Present

- **Autonomous Operations:** Manage job-site logistics, equipment operations, and daily resource provisioning, executing autonomous real-time decisions without direct managerial supervision.

LEADERSHIP & COMMUNITY

- **President, Spanish National Honor Society:** Organized community outreach initiatives and bi-weekly assemblies, leveraging bilingual skills to bridge English- and Spanish-speaking community groups.
- **Aviation Club Webmaster:** Built and optimized the club's web platform to support member recruitment and community airport events.
- **Sustained Community Volunteer:** Contributed 100+ volunteer hours maintaining trails and facilities across Veterans' Hall, Volunteer Trail Steward, and Trinidad Coastal Land Trust.